Woo-Jin Cho Kim

Personal Data

EMAIL: woojinchokimm@hotmail.com
HOMEPAGE: woojinchokimm.github.io
LINKEDIN: woojinchokim

Toolkit and Stack

Frameworks:	PyTorch, TensorFlow, Keras, Scikit
Data & Processing:	Pandas, NumPy, OpenCV, Albumentations, Timm
Deployment & Ops:	Docker, AWS SageMaker, ONNX, TFServing
Languages & Tools:	Python, Bash, Git, SQL, W&B

Work Experience

Current – Jan 2024 | **Deep Learning Scientist**, Ultromics — Oxford, UK
AI and Computer Vision Group
- Developed and deployed ML pipelines for image classification and sequence modeling on medical ultrasound data.
- Pioneered reinforcement learning strategy to reduce patient scan needs by up to 70% for cardiac disease diagnosis.
- Pretrained ultrasound foundation models using MAE and DINOv2, applying advanced augmentation strategies for improved generalization.
- Fine-tuned 🤗 foundation models for downstream tasks with LoRA and ExPLoRA.
- Delivered scalable, production-grade inference services using ONNX, Docker, and Sage-Maker, integrated into cloud-based clinical pipelines.

Jan 2024 – Apr 2022 | **Junior Deep Learning Scientist**, Ultromics — Oxford, UK
- Built modular medical data preprocessing pipelines for high-volume image ingestion, normalization, and augmentation to support development of ML models.
- Tuned model architectures and training schedules for FDA validation protocols.
- Worked with clinicians to validate model predictions via different attention modes.

Jan 2022 – Sept 2021 | **Research Intern**, Ultromics — Oxford, UK
- Evaluated segmentation models with a focus on uncertainty quantification to input variability in patient data.

Nov 2023 – Dec 2022 | **PhD Candidate**, KCL x Ultromics — London, UK
- Developed a novel cardiac volume quantification method for clinical ultrasound data.
- Now integrated into production-grade pipeline for real-time cardiac decision support.

Education

Aug 2023 – Sept 2017 | PhD in AI for Medical Image Computing
King's College London Dr. Pablo Lamata, Dr. Andrew King
Thesis topic: Improving echocardiographic diagnostic accuracy.

Sept 2017 – Sept 2016 | MSc in Computer Science (ML Specialization)
Imperial College London Dr. Panos Parpas

Sept 2015 – Sept 2012 | BEng in Biomedical Engineering
King's College London Dr. Oleg Aslanidi
Rosalind Franklin Prize for Respiratory Motion Modelling Project, MIUA 2015

Selected Publications

1. **Automated Detection of Apical Foreshortening in Echocardiography Using Statistical Shape Modelling,** Cho Kim W. J.*, Leeson P., Lamata P., et al. *UMB 2023*
2. **Beyond Simpson's Rule: Accounting for Orientation and Ellipticity Assumptions** Cho Kim W. J.*, Lewandowski A. J., King A. P., Leeson P., Lamata P., et al. *UMB 2022*
3. **BackMix: Mitigating Shortcut Learning in Echocardiography with Minimal Supervision** Bransby K., Beqiri A., Cho Kim W.J., Chartsias A., Gomez A., et al. *MICCAI 2024*
4. **Multi-Site Class-Incremental Learning with Weighted Experts in Echocardiography** Bransby K., Beqiri A., Cho Kim W.J.., Chartsias A., Gomez A., et al. *ASMUS 2024*